Anti-Anxiety Medications

Goal:

To provide the participants with knowledge, skills and positive attitude toward anti-anxiety medications.

Objectives

At the end of this presentation the participants will be able to:

- Identify anti-anxiety drugs
- Classify anti-anxiety drug types
- Know the mechanism of actions of anti-anxiety drugs
- Discuss the indications and contraindications of anti-anxiety drugs
- Identify the most common side effect of anti-anxiety drugs
- Determine the precaution of anti-anxiety drugs

Outlines:

- Definition of anti-anxiety drugs (anxiolytics)
- Types of anti-anxiety drugs
  - Benzodiazepines
  - Selective Serotonin Reuptake Inhibitor (SSRI)
  - Serotonin and Norepinephrine Reuptake Inhibitors (SNRIs)
  - Beta-blockers
- General health education

Definition of crisis:

Definition of anti-anxiety drugs (Anxiolytics):

Anxiolytics (also termed anti-anxiety or anti-panic drugs) are medications that are used to treat a health condition called anxiety that causes excessive worry and affects the ability of a person to function.

Types of anti-anxiety medications:

1) Benzodiazepines
2) Selective Serotonin Reuptake Inhibitors (SSRIs)
3) Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)
4) Beta-blockers

1- Benzodiazepines (BZDs)

Mechanism of Action:

- Benzodiazepines bind to specific sites on the GABA-A receptor complex, which increases the frequency of chloride channel opening, leading to hyperpolarization of the neuron and inhibition of neurotransmitter release. This enhances the inhibitory effects of GABA, resulting in the desired therapeutic effects.

List of commonly prescribed benzodiazepines:

1. Alprazolam (Xanax)
2. Diazepam (Valium)
3. Lorazepam (Ativan)
4. Clonazepam (Klonopin)

Indications of (BZDs):

- Anxiety disorders (generalized anxiety disorder, panic disorder, social anxiety disorder)
- Insomnia
- Seizure disorders
- Muscle spasms
- Alcohol withdrawal symptoms
- Preoperative sedation

Contraindications of (BZDs):

- Hypersensitivity to benzodiazepines or any of their components
- Acute narrow-angle glaucoma.
- Severe respiratory insufficiency.
- Sleep apnoea.
- Severe hepatic impairment.
- Pregnancy (especially during the first trimester) and lactation (due to potential risks to the foetus or new-born)

Side Effects of (BZDs).

- Sedation
- Drowsiness
- Dizziness
- Impaired coordination
- Memory impairment
- Confusion
- Tolerance and dependence with long-term use
- Withdrawal symptoms upon discontinuation (especially with abrupt cessation)

Precautions and Health Education for (BZDs):

- Avoid Central nervous system depressants (e.g., alcohol, opioids, and other sedatives/hypnotics): Increased risk of respiratory depression and sedation.
- Use caution when performing tasks requiring mental alertness or coordination, such as driving or operating machinery.
- Do not exceed prescribed doses or abruptly discontinue benzodiazepines without medical supervision to avoid withdrawal symptoms.

2- Selective Serotonin Reuptake Inhibitors (SSRIs)

Mechanism of action:

- Selectively inhibit the central nervous system neuronal uptake of serotonin
- It's thought that SSRIs work by increasing serotonin levels in the brain.
- Serotonin is usually reabsorbed by the nerve cells (known as "reuptake").
- SSRIs work by blocking ("inhibiting") reuptake, meaning more serotonin is available to pass further messages between nearby.

List of commonly prescribed (SSRIs):

1. Citalopram (Celexa)
2. Escitalopram (Lexapro)
3. Fluoxetine (Prozac)
4. Paroxetine (Paxil, Pexeva)
5. Sertraline (Zoloft)

Indications of (SSRIs):

1. Generalized anxiety disorder (GAD)
2. Obsessive-compulsive disorder (OCD)
3. Recurrent panic attacks
4. Social anxiety disorder (SAD)
5. Severe phobias, such as agoraphobia and social phobia
6. Post-traumatic stress disorder (PTSD)

Contraindications of (SSRIs):

1. Hypersensitivity to SSRIs
2. Concomitant use of SSRIs with, or within 14 days' use of, MAOIs
3. Hepatic or renal insufficiency
4. Pregnancy (first trimester) and lactation

Adverse Reactions and Side Effects of(SSRIs):

- Lethargy
- Drowsiness
- Anorexia
- Diarrhea
- Increased risk of suicidality in children and adolescents (black box warning)
- Sexual dysfunction
- Dry mouth
- Serotonin syndrome.

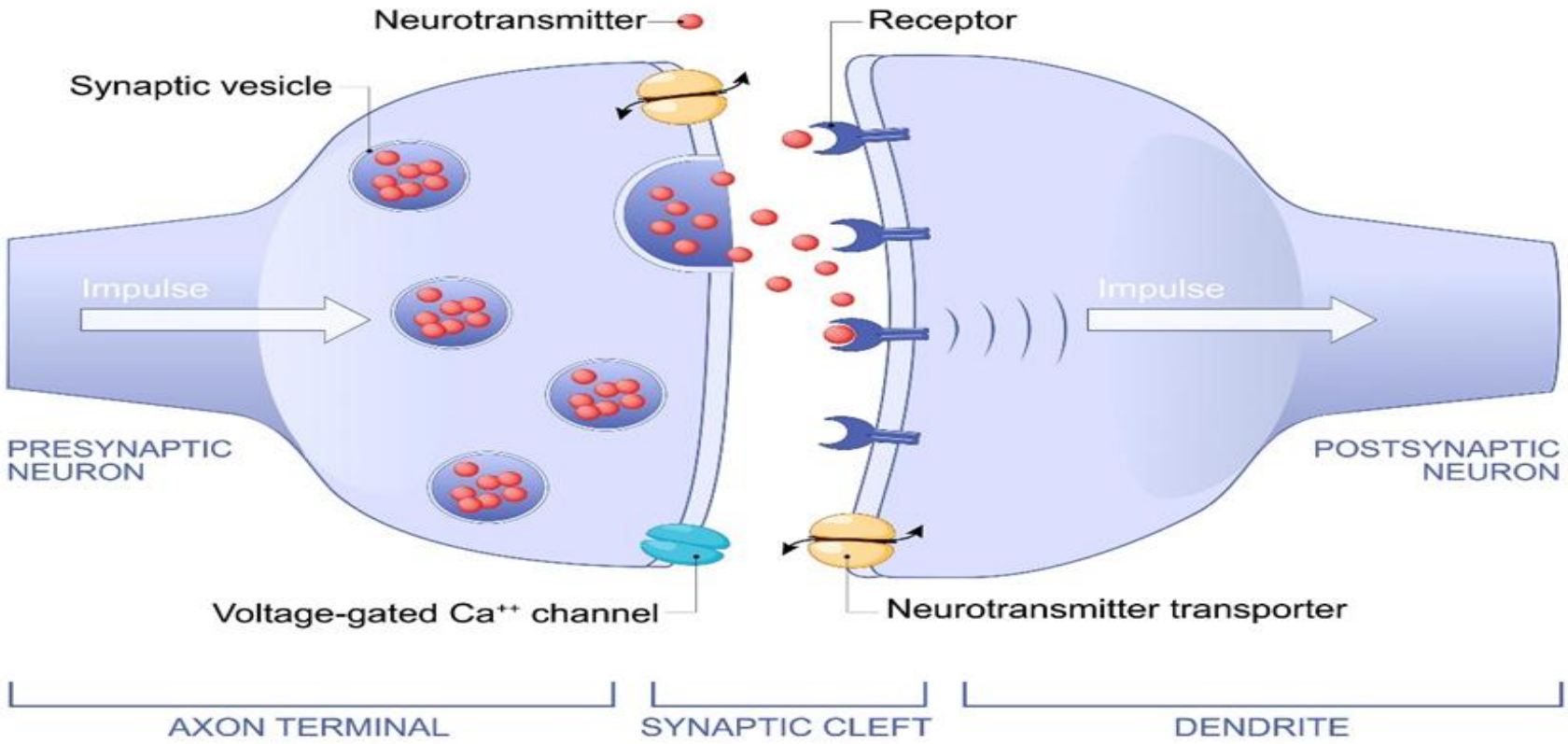
3- Serotonin and Norepinephrine Reuptake Inhibitors (SNRIs)

Mechanism of action of (SNRIs):

- SNRIs inhibit the reuptake of serotonin (5-HT) and norepinephrine (NE) by blocking their transporters, increasing the levels of these neurotransmitters in the synaptic cleft, and enhancing neurotransmission.

Neurotransmission

(the presynaptic neuron releases a neurotransmitter, which activates receptors on the postsynaptic neuron)



Indications of SNRIs :

1. generalized anxiety disorder (GAD)
2. Depression
3. Recurrent panic attacks
4. Social anxiety disorder (SAD)

Contraindications of (SNRIs):

- Hypersensitivity to the drug
- Concomitant (or within 14 days) use with MAOIs
- Severe renal or hepatic impairment
- Pregnancy and lactation
- Uncontrolled narrow-angle glaucoma

List of commonly prescribed of SNRIs :

- Duloxetine (Cymbalta)
- Venlafaxine (Effexor-XR)

4. Beta-blockers

- Drugs used for short-term treatment.
- Treatment of physical symptoms does not treat the root of the problem.
- Helps control heart rate.

List of commonly prescribed of Beta-blockers

- **Propranolol (Inderal)**

Thank you

Any question? !?